

Jihan Chowdhury

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EDUCATION

Toronto Metropolitan University

Bachelor of Engineering in Computer Engineering (Software Option), **GPA: 3.77**

Toronto, ON

Expected Apr. 2028

Relevant Coursework: Algorithms & Data Structures, Operating Systems, Database Systems, Software Architecture, Computer Organization & Architecture

TECHNICAL SKILLS

Languages: Python, Java, C++, TypeScript/JavaScript, SQL, C, Bash/Shell, VHDL, Assembly

Frameworks & Libraries: React, Next.js, Node.js, Express, Flask, Pandas, Drizzle, NextAuth, scikit-learn, Tailwind

Databases: PostgreSQL, Oracle SQL, MongoDB

Tools & Platforms: Git/GitHub, REST APIs, Docker, AWS, Azure, Linux/Unix, JUnit, SAP ERP

EXPERIENCE

Hitachi Rail

Feb. 2026 – Present

AI Intern (Software Engineering & Automation)

Toronto, ON

- Built a Python document-processing pipeline over **20–30 files/week** (PDF, XLSX, MSG) from SharePoint, extracting fields from non-searchable PDFs via an **Azure AI endpoint** with pages sent as base64-encoded images.
- Designed **tiered failure handling**: schema-validated model output, escalation to a larger-context model on parse failure, then log-and-skip so one malformed file never halted a batch.
- Engineered an **OCR-assisted redaction pipeline** for scanned PDFs with no text layer: image-level edits driven by word-level pixel coordinates, verified by re-OCRing the output. Cut **3 hours per 10-document batch**.
- Resolved records to their correct SAP cost object through **deterministic multi-hop lookups** across linked tables, replacing manual per-record coding.
- Deployed as a **scheduled job on an Azure VM** and handed off to IT to run in their environment, with externalized config and per-file logging.

PROJECTS

League Insights AI | Python, Flask, PostgreSQL, AWS, Next.js, React, TypeScript

Nov. 2025 – Jan. 2026

- Architected a **1,700+ line asynchronous Flask backend** pulling **1,000+ match records per player** across **200+ users/month** from a rate-limited API, using batched concurrent requests with **bounded concurrency** and exponential backoff to sustain throughput.
- Modeled a **40+ column PostgreSQL schema** with indexed lookups and chained **Alembic migrations**, writing in batches rather than per-row ORM inserts.
- Derived **windowed performance metrics** from raw match timeline frames; integrated AWS Bedrock for report generation and **led a 3-person team** defining backend interfaces.

Issue Tracker | Next.js 15, TypeScript, PostgreSQL, Drizzle ORM, NextAuth

Oct. 2025

- Built a full-stack issue tracker on the **Next.js App Router**, fetching through **React Server Components** and mutating through **Server Actions**, with session-gated writes and bcrypt-hashed credential auth.
- Modeled a **3-table Drizzle schema** with FK constraints and versioned migrations; resolved author data via server-side joins and paired **per-route ISR** with targeted cache invalidation so writes surfaced immediately.

E-Commerce DBMS | Oracle SQL, PHP, Unix

Sep. 2025 – Nov. 2025

- Designed a **normalized 11-table Oracle schema** (3NF/BCNF) modeling an **ISA specialization hierarchy** and an **M:N junction table**, with named FK and domain CHECK constraints.
- Wrote analytical SQL (**GROUP BY/HAVING**, **EXISTS** correlated subqueries, **MINUS/UNION** set operations) surfaced through reusable **views**, and automated the drop/create/populate/query lifecycle with **shell scripts** piping SQL into sqlplus behind a menu driver.

8-Bit Processor Datapath | VHDL, Quartus, FPGA

Dec. 2025

- Composed an **8-bit datapath in VHDL** from discrete modules – clocked registers with asynchronous reset, a **3-to-8 one-hot decoder** generating opcode select lines, and a **Moore finite state machine** driving sequencing.
- Implemented **multiple ALU variants** over an 8-bit one-hot opcode space (arithmetic, bitwise logic, nibble swap, bit reversal, compare); verified against **Quartus waveforms** and on FPGA hardware.